

C.V RAMAN GLOBAL UNIVERSITY

MID-SEMESTER EXAMINATION 2025-26(EVEN)

B.Tech

Subject :- DATA STRUCTURE

CSE

Sub. Code: CS102

er: 2nd

GROUP: ALL Groups

Time: 1 Hour
Max marks: 20

figures in the right-hand margin indicate marks.

PART A

Answer ALL questions:

- a Evaluate the prefix expression: $+ - 2 7 * 8 / 4 12$ [5x2]
- b Given a stack of size 5, perform the following operations: PUSH 10, PUSH 20, PUSH 30, POP, PUSH 40, POP. Show the stack after each operation. 1, 20
- c Define Double Ended Queue (Deque). List the different types of deque. / * + ABC * + DEF
- d Convert the following infix expression into prefix form using the parenthesis method: $((A + B) * C) / ((D + E) * F)$
- e Explain how a node is represented in a linked list and write the C statement used for dynamic memory allocation of a node.

PART-B(Answer Any Two)

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(2 x5)

Convert the following infix expression into postfix expression using stack.

$A - (B / C + (D / E * F) / G) * H$

DE * F

Write the C code / pseudocode to implement following operations in a single linked list.

- Insert a node after a specific value.
- Insert a node at a specific position.

Write the C code / pseudocode to implement insertion and deletion operation of a circular queue using array.

Q.1 a
Q.1 b

CO1
CO1
CO1

BL-2
BL-2
BL-1

Q.1 d
Q.1 e
Q.3

CO1
CO2
CO2

BL-3
BL-2
BL-3